

Session 5 · Deck 5.B4

Finite Elements

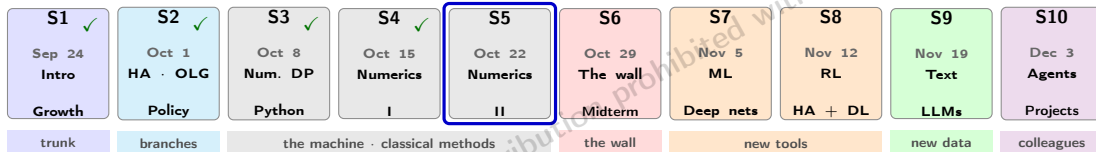
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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are

you are here: 5.B4 ▼



Build it on paper → solve it on the machine → push it to the frontier with AI

Block B: 5.B1 the residual · 5.B2 choosing a basis · 5.B3 Chebyshev · **5.B4** finite elements · 5.B5 determining the coefficients · 5.B6 the worked example.

The Opposite Choice

A spectral basis is **global**: every ψ_i is non-zero almost everywhere, so every coefficient affects the answer everywhere. That is what makes it converge geometrically on smooth functions — and what makes a single kink contaminate the whole domain.

A **finite element** basis is local. Chop the state space into elements, put a simple function on each, and let each basis function be non-zero only on its own small neighbourhood.

Then a kink is not a problem; it is a place to put an element boundary.

Quant_Macro Lec. 9, "Finite Element Method (FEM)".

Where FEM comes from

Engineering, for PDEs on awkward domains — irregular boundaries, corners, complex geometry. The mesh can be shaped to fit the problem.

In macroeconomics the “irregular boundary” is usually a constraint: $a' \geq -\underline{a}$, or $i_t \geq 0$. Put a node exactly there.

The Recipe

1. **Partition** the state space into elements — intervals in one dimension, triangles or rectangles in more.
2. **Choose a basis on each element**, typically piecewise polynomial, and typically 1 **at one node** and 0 **at all the others**. Summed, they form a **partition of unity**.
3. **Substitute** the approximant into the equilibrium conditions and integrate over each element, often by parts.
4. **Assemble** the element equations into one global algebraic system.
5. **Solve** for the nodal coefficients, and **paste** the pieces together so the result is continuous.

Steps 1–2 are the basis choice of 5.B2; steps 3–4 are the projection of 5.B5. **Nothing in the framework changed** — only $\{\psi_i\}$.

“Paste it together to ensure continuity” is doing real work: the elements are solved jointly, not one at a time.

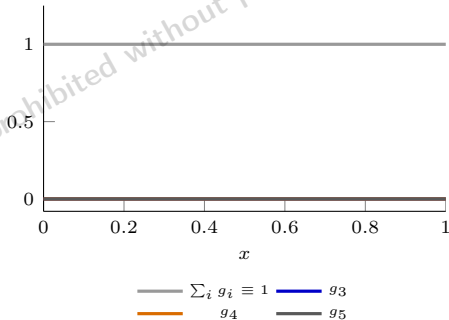
The Simplest Basis: Tent Functions

Elements are small, so a linear basis is usually enough:

$$g_i(x) = \begin{cases} \frac{x - x_{i-1}}{x_i - x_{i-1}} & x \in [x_{i-1}, x_i] \\ \frac{x_{i+1} - x}{x_{i+1} - x_i} & x \in [x_i, x_{i+1}] \\ 0 & \text{elsewhere} \end{cases}$$

Each g_i is 1 at its own node, 0 at every other node, and non-zero only on $[x_{i-1}, x_{i+1}]$.

Then $d^n(x) = \sum_i \theta_i g_i(x)$ and θ_i is simply **the value of the policy at node i** — the coefficients are interpretable in a way Chebyshev coefficients never are.



Quant_Macro Lec. 9, "Structure". Nine nodes on $[0, 1]$; measured $\max_x |\sum_i g_i(x) - 1| = 0$.

Check the Partition of Unity Before You Solve Anything

The tent function has two branches and it is easy to get the second one's sign wrong. Write

$$\frac{x_{i-1} - x}{x_i - x_{i-1}} \quad \text{for } x \in [x_i, x_{i+1}]$$

instead of $(x_{i+1} - x)/(x_{i+1} - x_i)$, and evaluate just to the right of the node:

$$g_i(x_i^+) = \frac{x_{i-1} - x_i}{x_i - x_{i-1}} = -1.$$

A basis function that is +1 arriving at its node from the left and -1 leaving it to the right is discontinuous and negative. Summed over nine nodes it is off by as much as 2; the correct basis is off by 0.

Ledger fem: $\max_x |\sum_i g_i(x) - 1| = 0$ for the correct basis and 2.0 with the sign flipped; value just past the node = -1.

Why it matters

The partition-of-unity property is what makes θ_i the policy value at node i , and what makes the assembled approximation continuous. Lose it and every interpretation on the previous slide fails — silently, because the solver still returns coefficients.

One line of code separates the two: sum the basis over a fine grid and compare to 1. Do it before solving anything, on every basis you write yourself.

Why the System Is Sparse

Basis function g_i is non-zero only on $[x_{i-1}, x_{i+1}]$. So in any integral of the form $\int g_i g_j$ or $\int g_i R$, the integrand is zero unless g_i and g_j **share an element**.

The assembled system $A\theta = b$ therefore has non-zeros only on the diagonal and its immediate neighbours: it is **banded**, with bandwidth 1.

Measured, on nine nodes: **25 non-zero entries out of 81**, a density of 30.9% — and the density falls like $3/n$ as the mesh is refined. At 500 nodes it is under 1%.

What sparsity buys

A dense solve is $O(n^3)$; a banded one is $O(n)$. That is the trade for FEM's slow convergence: many more coefficients, each one far cheaper.

And you never actually invert it — you factor it, which for a banded matrix is linear in the number of nodes.

Quant_Macro Lec. 9, "Why Does This Lead to a Sparse Matrix?". Ledger fem: 25/81 non-zeros, bandwidth 1.

Three Ways to Refine

h-refinement	Subdivide every element, improving resolution <i>uniformly</i> over the domain. The obvious move, and usually the wasteful one.
r-refinement	Subdivide only where the solution is strongly nonlinear. The mesh follows the economics — more elements near a binding constraint, few where the policy is nearly linear.
p-refinement	Raise the polynomial order <i>inside</i> each element. Push it far enough and you have a hybrid of finite and spectral methods: spectral elements .

The design space is continuous

One element with a degree- n basis is a spectral method. n elements with a degree-1 basis is FEM. Everything between is available.

r-refinement is the one that pays in macro, and it is the same instinct as the log-spaced asset grid of 3.A2: put the points where the curvature is.

Finite Elements Against Smolyak

	Finite elements	Smolyak
Basis	piecewise polynomial, local ; the solution may differ element by element	global Chebyshev or splines on a sparse index set
Grid	a mesh , which can follow irregular boundaries and be adapted locally	tensor product of 1D grids, thinned by the sparse rule
Strength	low to moderate dimension with awkward domains or kinks	moderate to high dimension with smooth solutions
Weakness	heavy in high dimension — the mesh itself is $O(n^d)$	less natural for irregular domains and non-smooth solutions

They fail in **opposite** directions, which is the useful thing about the comparison: dimension kills FEM, and non-smoothness kills Smolyak. Knowing which one you face is the choice.

Takeaways

- Finite elements are the **local** basis: chop the domain, put a simple function on each piece. A kink stops being a problem and becomes a place to put an element boundary.
- The tent basis is 1 at its own node and 0 at every other, so the coefficients **are** the policy values at the nodes — directly interpretable, unlike Chebyshev coefficients.
- **Check the partition of unity before solving anything.** One sign slip in the tent formula gives a basis worth -1 just past its own node, off by 2 when summed; the correct one is exact to 0, and one line of code tells them apart.
- Compact support makes the assembled system **banded**: measured 25 non-zeros out of 81 on nine nodes, and the density falls like $3/n$. A banded solve is $O(n)$, not $O(n^3)$.
- **h-, r-, p-refinement**: more elements, better-placed elements, or higher order within elements. Push p far enough and you have spectral elements — the two families are ends of one design space.
- FEM and Smolyak fail in **opposite** directions: dimension defeats FEM, non-smoothness defeats Smolyak.

Bridge: What Does “Small” Mean?

Three decks have now been spent on the first of 5.B1's two questions — which basis. Spectral or local, Chebyshev or tents, dense or sparse.

The second question has been deferred every time it came up: having written $R(x; \theta)$, what exactly do we do to it? “Make it small” is not an algorithm, and the choices are not equivalent.

Next (5.B5)

Weighted residuals, and the six answers: least squares, subdomain, moments, collocation, orthogonal collocation, and Galerkin. All six run on the same model, with the same basis — and the results compared.